Dynamical Calibration of a Metallicity-Dependent Stellar Mass–Luminosity Relation from Gaia Wide Binaries

Model with selection-corrected dynamical likelihood: 3 October 2026

1 Scientific objective and modular architecture

We use modular inference. Stage 1a uses within-binary metallicity differences to estimate the relative JCAPS calibration. Conditional on that estimate, Stage 1b combines the component JCAPS measurements with anchor stars' colour–magnitude diagram (CMD) and a hierarchical population distribution to infer a probability distribution for the shared metallicity of every binary. Stage 2 treats these distributions as fixed inputs, predicts the two component masses at every metallicity, and evaluates the wide-binary dynamical likelihood, corrected for the selection of the sample by the significance of the proper-motion difference. The architecture is summarized in figure 1.

2 Observables, latent quantities, and conventions

Binary systems are indexed by j, their two components by $k \in \{1, 2\}$, and metallicity nodes by q. A hat denotes an observed quantity. The symbol $\log_{10}$ is reserved for base-10 stellar-mass logarithms; unqualified log denotes the natural logarithm. "Dex" denotes an interval in a base-10 logarithm. The main quantities are listed in table 1.

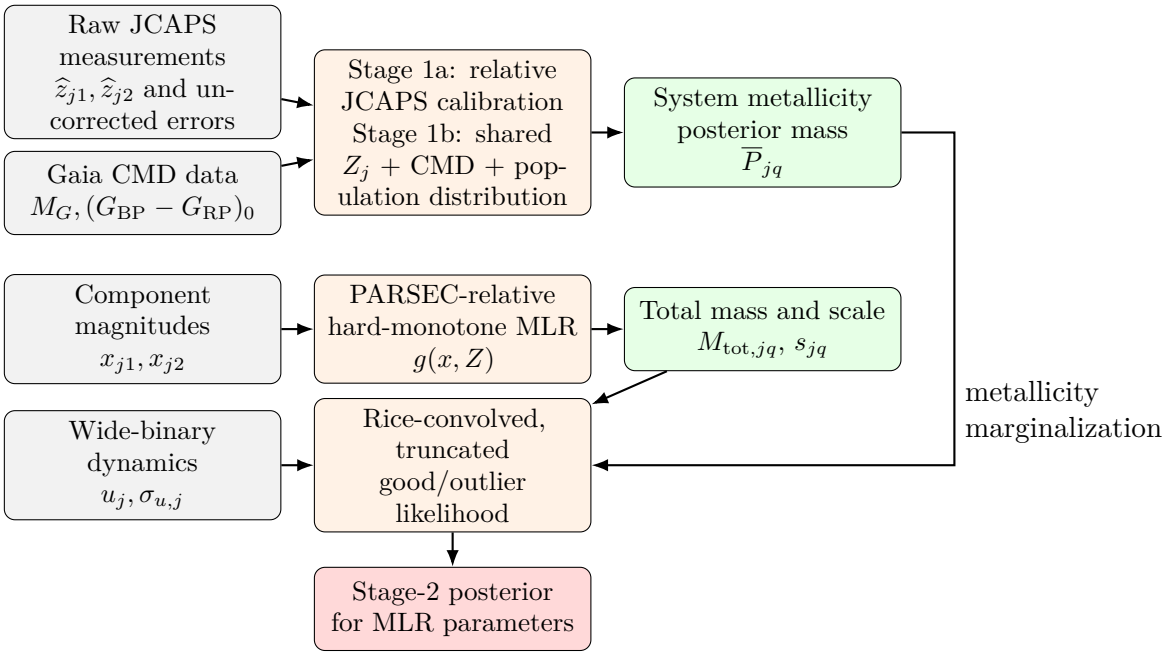


Figure 1: Scientific workflow. Stage 1a estimates the relative JCAPS calibration from within-binary differences. Conditional on those fixed estimates, Stage 1b returns a posterior probability mass over each shared system metallicity. Stage 2 propagates this uncertainty into the dynamical MLR fit but does not update either Stage 1 module. This one-way flow implements the modular cut.

Table 1: Principal observables and latent or derived quantities.

Symbol	Meaning	Unit or range
$\mathcal{D}_{1a}$	Within-binary JCAPS differences used to estimate relative calibration parameters	data set
$\mathcal{D}_{1b}$	JCAPS and Gaia CMD data used for absolute calibration and system posteriors	data set
$\mathcal{D}_2$	Stage-2 component-magnitude and wide-binary dynamical data, with $\overline{P}_{jq}$ supplied from Stage 1b	data set
x_{jk}	Gaia absolute magnitude of component k	mag
C_{jk}	Dereddened colour $(G_{\rm BP} - G_{\rm RP})_0$	mag
$\widehat{z}_{jk}$	Raw JCAPS XP metallicity measurement	dex
$\sigma_{z,jk}$	Reported, uncorrected JCAPS uncertainty	dex
Z_j	Shared true system metallicity on the PARSEC scale	dex
$\overline{P}_{jq}$	Stage-1b posterior probability mass at Z_q	$[0, 1]$
$g(x, Z)$	Final $\log_{10}(M/M_\odot)$ surface	dex
u_j	Observed $\Delta v_{\tan}\sqrt{r_\perp}$ amplitude	$\mathrm{km\,s^{-1}}\sqrt{\mathrm{AU}}$
$\sigma_{u,j}$	Component-wise Gaussian error scale entering the Rice model	$\mathrm{km\,s^{-1}}\sqrt{\mathrm{AU}}$
s_{jq}	Dynamical mass scale $\sqrt{M_{{\rm tot},jq}/M_\odot}$	dimensionless
c	Threshold of the catalogue cut $u_j/\sigma_{u,j} > c$ $(c = 3)$	dimensionless
$\mathcal{A}_j$	Probability that pair j passes the cut	$[0, 1]$

The metallicity inputs are strictly `feh_jcaps_1`, `feh_jcaps_2`, `jc_sigma_m_h_1`, and `jc_sigma_m_h_2`. JCAPS infers stellar labels by inverting a forward model of Gaia XP spectra; the underlying framework and its training strategy are described by Li et al. (2026). The names above refer to columns in the analysis table and are not asserted to be published JCAPS catalogue field names. No binary-corrected, fitted, calibrated, or inflated metallicity column is used. Such columns already incorporate the equal-composition assumption that appears in our likelihood; reusing them would count that information twice and make the calibration circular.

3 Stage 1: relative and absolute metallicity calibration

3.1 Stage 1a: Relative JCAPS calibration

Before introducing the PARSEC CMD, the relative behavior of JCAPS is calibrated from the difference between the two stars in each binary. The adopted observation model assigns one independent Student-t state to each star,

$$\widehat{z}_{jk} \mid Z_j \sim \mathcal{T}_\nu\Big[Z_j + \delta_z + b_G(x_{jk}), \sqrt{\sigma^2_{z,jk} + s_z^2}\Big] = \mathcal{L}^{\mathrm{spec}}_{jk}(Z_j), \tag{1}$$

where the second argument is the location and the third is the scale. The function $b_G(x)$ is a magnitude-dependent *relative* bias, s_z is one common additional scale per star (hence a contribution $2s_z^2$ to a binary difference variance), and ν is the shared degrees of freedom. Conditional on Z_j, the two component measurements are independent.

3.2 Stage 1b: Absolute calibration and system metallicity posteriors

3.2.1 CMD likelihood

In principle, we can make use of any set of anchor points whose metallicties are robust and cover a wide enough range to get the absolute calibration.

Here's an example when we directly use the PARSEC model points of warm main-sequence stars. The PARSEC surface supplies predicted colour $C_{\mathrm{P}}(x, Z)$ and effective temperature $T_{\mathrm{eff,P}}(x, Z)$ (Bressan et al., 2012). The repository records the age, metallicity, and magnitude coverage of the local PARSEC table, but not enough provenance to identify a more specific PARSEC release or Gaia passband realization; we therefore make no more specific version claim. The formal colour uncertainty is propagated from the BP and RP signal-to-noise ratios,

$$\sigma_{C,jk} = \frac{2.5}{\ln 10}\left(\mathrm{SNR}^{-2}_{\mathrm{BP,jk}} + \mathrm{SNR}^{-2}_{\mathrm{RP,jk}}\right)^{1/2}. \tag{2}$$

For an eligible component, the CMD likelihood is

$$C_{jk} \mid Z_j \sim \mathcal{T}_4\Big[C_{\mathrm{P}}(x_{jk}, Z_j), \sqrt{\sigma^2_{C,jk} + s_C^2}\Big] = \mathcal{L}^{\mathrm{CMD}}_{jk}(Z_j). \tag{3}$$

The colour zero point is fixed to zero and $s_C \sim \mathrm{HalfNormal}(0.1\ \mathrm{mag})$. To avoid allowing membership in the CMD term to change with a proposed metallicity, eligibility is fixed over the full latent-metallicity domain:

$$m_{jk} = \mathbb{I}\left[\min_{q\in\mathcal{Q}} T_{\mathrm{eff,P}}(x_{jk}, Z_q) \geq 4000\ \mathrm{K}\right]. \tag{4}$$

If $m_{jk} = 0$, that star still contributes its JCAPS likelihood but not its CMD likelihood. To construct the fixed CMD reference surface, we selected PARSEC rows with $9.3 < \log_{10}(\mathrm{age/yr}) < 10.0$ and $3 < G_{\mathrm{mag}} < 15$. At fixed source-table G_{mag} and metallicity, rows from the selected ages were averaged separately in colour and $\log T_{\mathrm{eff}}$, so age is not a latent variable in this analysis. Colour was represented by an up-to-cubic smoothing spline with smoothing parameter 0.001;

$\log T_{\rm eff}$ was linearly interpolated and then exponentiated before tabulation. The colour and linear-$T_{\rm eff}$ tables were evaluated bilinearly in (M_G, Z) on a 500-point magnitude grid spanning $3.5 \le M_G \le 13.5$. This CMD construction is distinct from the 501×81 (M_G, Z) grid used only to fit the MLR's fixed PARSEC mass projection.

The sample contains 14,876 binaries (29,752 stars). The 4000-K rule admits 12,167 stars; 8,247 systems have at least one eligible component and 3,920 have two. These counts describe CMD coverage, not evidence that the cool-star PARSEC colours are correct.

3.2.2 System posterior and the cut object

The sampled Stage-1a and Stage-1b global parameters are $\boldsymbol{\psi} = (b_{\rm G}, s_z, \delta_z, s_C)$. The unnormalized metallicity density for one system is

$$\widetilde{p}_j(Z, \boldsymbol{\psi} \mid \mathcal{D}) = p_{\rm pop}(Z) p(\psi) \prod_{k=1}^{2} \mathcal{L}_{jk}^{\rm spec}(D_{\rm 1a} \mid Z, \boldsymbol{\psi}) \left[\mathcal{L}_{jk}^{\rm CMD}(D_{\rm 1b} \mid \boldsymbol{\psi})\right]^{m_{jk}}. \tag{5}$$

The global parameters are inferred from the product of the system-level marginal evidences, rather than from separately normalized system posteriors.

The normalized system posterior is

$$p_j(Z \mid \mathcal{D}) = \frac{\widetilde{p}_j(Z, \boldsymbol{\psi} \mid \mathcal{D})}{\int \widetilde{p}_j(Z, \boldsymbol{\psi} \mid \mathcal{D})\, dZ}. \tag{6}$$

4 Stage 2: a PARSEC-relative hard-monotone MLR

4.1 Baseline projection and scientific correction

Let the original PARSEC mass surface be

$$g_{\rm P}(x, Z) = \log_{10}\left[\frac{M_{\rm P}(x, Z)}{M_\odot}\right]. \tag{7}$$

The finite MLR basis cannot reproduce this tabulated surface exactly. The projected PARSEC mass surface onto the same tensor-product B-spline basis used by the fitted model is represented as $g_{\rm P}^*(x, Z)$,

$$g_{\rm P}^*(x, Z) = \sum_{i=1}^{K_x} \sum_{h=1}^{K_Z} \Pi_{ih} B_i(x) C_h(Z), \tag{8}$$

where Π is fixed. The fitted correction is

$$\Delta_*(x, Z) = \sum_{i,h} D_{ih} B_i(x) C_h(Z), \qquad \Theta_{ih} = \Pi_{ih} + D_{ih}, \tag{9}$$

and hence the fitted mass surface is

$$g(x, Z) = \sum_{i,h} \Theta_{ih} B_i(x) C_h(Z) = g_{\rm P}^*(x, Z) + \Delta_*(x, Z). \tag{10}$$

The quantity reported scientifically relative to the original PARSEC table is not Δ_* alone but

$$\Delta_{\rm P}(x, Z) = g(x, Z) - g_{\rm P}(x, Z) = \Delta_*(x, Z) + \epsilon_{\rm P}(x, Z), \quad \epsilon_{\rm P} = g_{\rm P}^* - g_{\rm P}. \tag{11}$$

where $\epsilon_{\rm P}$ is the deterministic projection error, the RMS and maximum absolute values of $\epsilon_{\rm P}$ are 0.002 and 0.019 dex, respectively.

The adopted basis has $K_x \times K_Z = 8 \times 4$ cubic B-splines (de Boor, 2001). Both knot vectors are open and clamped:

$$\boldsymbol{\kappa}_x = (3.5, 3.5, 3.5, 3.5, 5.5, 7.5, 9.5, 11.5, 13.5, 13.5, 13.5, 13.5)\ \text{mag}, \tag{12}$$

$$\boldsymbol{\kappa}_Z = (-1.0, -1.0, -1.0, -1.0, 0.6, 0.6, 0.6, 0.6)\ \text{dex}. \tag{13}$$

4.2 Hard monotonicity of the total mass surface

At fixed metallicity, increasing M_G means a fainter star and must not increase its mass. At fixed magnitude, the adopted physical assumption is that increasing metallicity must not decrease the mass. We therefore require

$$\Theta_{i+1,h} \leq \Theta_{ih}, \qquad \Theta_{i,h+1} \geq \Theta_{ih}. \tag{14}$$

Because B-spline derivatives are non-negative lower-degree basis functions weighted by adjacent control-coefficient differences, these inequalities imply throughout the continuous domain

$$\frac{\partial g}{\partial x} \leq 0, \qquad \frac{\partial g}{\partial Z} \geq 0. \tag{15}$$

The constraints apply to the *total* coefficient surface $\Theta = \Pi + D$, not to the correction D alone. The correction is consequently free to increase or decrease in either coordinate. We also impose no sign constraint on its mixed differences; doing so would add an unsupported assumption about how the magnitude slope changes with metallicity. The constructive parameterization that enforces equation (14) is given in appendix C.

5 Wide-binary dynamical likelihood

5.1 Mass-scaled velocity statistic

In binary orbital motion, the total mass $m_{\rm tot} = m_1 + m_2$, the orbital separation a, the true anomaly θ and the eccentricity e set the instantaneous relative speed,

$$v_{\rm rel} = \sqrt{\frac{G\, m_{\rm tot}}{a}}\, f(\theta, e), \tag{16}$$

where $f(\theta, e)$ describes the phase dependence along the ellipse. At pericentre ($\theta = 0$) and apocentre ($\theta = \pi$),

$$v_{\rm rel,peri} = \sqrt{\frac{G\, m_{\rm tot}}{a}}\sqrt{\frac{1+e}{1-e}}, \qquad v_{\rm rel,apo} = \sqrt{\frac{G\, m_{\rm tot}}{a}}\sqrt{\frac{1-e}{1+e}}. \tag{17}$$

We use the observable

$$u := v\sqrt{s}, \tag{18}$$

where v is the projected relative speed and s is the projected separation. Since $v \propto a^{-1/2}$ and $s \propto a$, the semi-major axis cancels in u, and $u \propto \sqrt{m_{\rm tot}/M_\odot}$ with a proportionality that depends only on e, the orbital phase and the orientation. Defining the mass-scaled statistic $\widetilde{u} \equiv u/\sqrt{m_{\rm tot}/M_\odot}$, these nuisance parameters are marginalized once in a simulation to obtain a distribution $p(\widetilde{u})$ that does not depend on the mass. For a noiseless measurement the likelihood is then

$$p(u \mid m_{\rm tot}) = \frac{1}{\sqrt{m_{\rm tot}/M_\odot}}\, p\!\left(\frac{u}{\sqrt{m_{\rm tot}/M_\odot}}\right). \tag{19}$$

In terms of the mass scale $s_{jq} = \sqrt{M_{{\rm tot},jq}/M_\odot}$ of table 1, with $M_{{\rm tot},jq} = M_{j1q} + M_{j2q}$ and $M_{jkq} = M_\odot 10^{g(x_{jk}, Z_q)}$, the mass dependence of pair j at metallicity node Z_q enters only through s_{jq}.

5.2 Rice measurement model and mixture likelihood

The observed u_j is the magnitude of a two-dimensional velocity difference. For independent Gaussian errors of common scale $\sigma_{u,j}$ on its two components, u_j given the latent speed v follows the Rice distribution (Rice, 1945),

$$R(u_j \mid v, \sigma_{u,j}) = \frac{u_j}{\sigma_{u,j}^2} \exp\left[-\frac{u_j^2 + v^2}{2\sigma_{u,j}^2}\right] I_0\left(\frac{u_j v}{\sigma_{u,j}^2}\right), \qquad u_j \geq 0, \tag{20}$$

where I_0 is the zeroth-order modified Bessel function. The scale is $\sigma_{u,j} = u_j/(\Delta\mu/\sigma_{\Delta\mu})_j$, where $\Delta\mu$ is the proper-motion difference of the pair and $\sigma_{\Delta\mu}$ its uncertainty.

The good component has the density of the functional form of Hwang et al. (2024),

$$p(\widetilde{u}) = \frac{1}{N}\, \widetilde{u} \exp\left[-B\widetilde{u}^2 - \exp\left(\frac{\widetilde{u} - u_c}{C}\right)\right] \mathbb{I}(0 < \widetilde{u} \leq 80), \tag{21}$$

with $B = 2.5050 \times 10^{-3}$, $u_c = 37.2461$ and $C = 2.4500$ in the units of $\widetilde{u}$, and N chosen so that p integrates to one on $(0, 80]$. These constants are a least-squares fit of equation (21) to the distribution of $\widetilde{u}$ from a Monte Carlo of Keplerian orbits with eccentricity distribution $p(e) \propto e^{\alpha_e}$, uniform mean anomaly and isotropic orientation, in which each simulated pair is weighted so that its joint distribution of projected separation and distance matches the sample. We adopt $\alpha_e = 1.23$: the separation-dependent index $\alpha(w) = 1.25 \tanh[(w - 1.87)/0.88] + 0.12$, $w = \log_{10}(s/\mathrm{AU})$, of Hwang et al. (2022), measured from the angle between the separation and relative-velocity vectors and therefore independent of stellar masses, is evaluated at the projected separation of each pair, one eccentricity is drawn per pair, and a single power law is fitted to the pooled values by maximum likelihood. Because the catalogue selection acts on the observed (s, d), and $\widetilde{u}$ does not depend on the semi-major axis or the total mass, this weighting reproduces the angular-resolution limit without a separate θ cut. The outlier component is a Gaussian of mean 40 and standard deviation $13\,\mathrm{km\,s^{-1}}\sqrt{\mathrm{AU}}$, truncated and normalized on $[0, 80]\,\mathrm{km\,s^{-1}}\sqrt{\mathrm{AU}}$ and defined directly in the latent speed v,

$$q(v) = \frac{\phi[(v - 40)/13]}{13\{\Phi[(80 - 40)/13] - \Phi(-40/13)\}} \mathbb{I}(0 \leq v \leq 80), \tag{22}$$

where ϕ and Φ are the standard normal density and distribution function. The outlier component therefore does not depend on the pair mass. The mixture likelihood of pair j before accounting for selection is

$$\mathcal{L}_j(s) = (1 - f_{\mathrm{outlier}}) \int_0^{80} p(\widetilde{u})\, R(u_j \mid s\widetilde{u}, \sigma_{u,j})\, \mathrm{d}\widetilde{u} + f_{\mathrm{outlier}} \int_0^{80} q(v)\, R(u_j \mid v, \sigma_{u,j})\, \mathrm{d}v. \tag{23}$$

The first integral carries the Jacobian $1/s$ of the change of variables from v to $\widetilde{u}$. The two component shapes are fixed and f_{outlier} is a free parameter.

5.3 Truncation by the significance cut

The catalogue retains only pairs whose proper-motion difference is detected at more than $c = 3$ times its uncertainty. Because $\sigma_{u,j} = u_j/(\Delta\mu/\sigma_{\Delta\mu})_j$, this cut is a threshold on the observed statistic,

$$u_j > c\, \sigma_{u,j}, \tag{24}$$

with $\sigma_{u,j}$ fixed for each pair by the catalogue astrometric uncertainties, the parallax and the projected separation. Pairs with small latent speed and large $\sigma_{u,j}$ are preferentially removed, so the observed u_j are not a draw from equation (23). The density of u_j conditional on passing the cut is

$$\mathcal{L}_j^{\mathrm{sel}}(s) = \frac{\mathcal{L}_j(s)}{\mathcal{A}_j(s)}, \tag{25}$$

where $\mathcal{A}_j(s)$ is the probability that a pair with mass scale s satisfies equation (24).

For a latent speed v, the Rice distribution gives the probability of passing the cut as the Marcum Q function,

$$\Pr(u > c\,\sigma_{u,j} \mid v) = Q_1\left(\frac{v}{\sigma_{u,j}},\, c\right). \tag{26}$$

Averaging over the two mixture components,

$$\mathcal{A}_j(s) = (1 - f_{\rm outlier})\,\mathcal{A}_j^{\rm good}(s) + f_{\rm outlier}\,\mathcal{A}_j^{\rm out}, \tag{27}$$

with

$$\mathcal{A}_j^{\rm good}(s) = \int_0^{80} p(\widetilde{u})\,Q_1\left(\frac{s\widetilde{u}}{\sigma_{u,j}},\, c\right)\,{\rm d}\widetilde{u}, \qquad \mathcal{A}_j^{\rm out} = \int_0^{80} q(v)\,Q_1\left(\frac{v}{\sigma_{u,j}},\, c\right)\,{\rm d}v. \tag{28}$$

The pass probability of the outlier component does not depend on the mass, and $\mathcal{A}_j^{\rm good}$ depends on the mass and the uncertainty only through $s/\sigma_{u,j}$. It increases with s and tends to one when $s\widetilde{u} \gg c\,\sigma_{u,j}$ over the support of $p(\widetilde{u})$. Dividing by $\mathcal{A}_j$ therefore removes the probability that the untruncated model places below the cut, where no pair is observed; this probability is larger at low mass, so the correction lowers the inferred masses relative to equation (23). The fraction of outliers in the selected sample is $f_{\rm outlier}\mathcal{A}_j^{\rm out}/\mathcal{A}_j$, which differs from $f_{\rm outlier}$; the parameter $f_{\rm outlier}$ is the mixing fraction before selection.

5.4 Marginalization over metallicity

Selection acts on the true mass of the pair, so the factor $\mathcal{A}_j$ is evaluated at each metallicity node before the sum over the Stage-1b weights,

$$\mathcal{L}_j^{\rm sel}(\Theta) = \sum_q \overline{P}_{jq}\,\frac{\mathcal{L}_j(s_{jq})}{\mathcal{A}_j(s_{jq})}, \tag{29}$$

where s_{jq} is determined by the MLR parameters Θ through equation (10). The dynamical term of the Stage-2 log-posterior is $\sum_j \ln \mathcal{L}_j^{\rm sel}$.

5.5 Fixed velocity distribution

The function $p(\widetilde{u})$ describes the distribution of the mass-scaled statistic before selection, determined by eccentricity, orbital phase and orientation. With the cut modelled in equation (25), its shape is held fixed and is not adjusted to absorb the selection. Its upper edge is set by energy conservation: for a bound orbit $u^2 \le v^2 r < 2Gm_{\rm tot}$, hence

$$\widetilde{u} < \sqrt{2GM_\odot/{\rm AU}} = 42.12\,{\rm km\,s^{-1}}\sqrt{\rm AU}, \tag{30}$$

independently of the eccentricity distribution. This edge fixes the mass scale of the velocity distribution.

5.6 Numerical evaluation

The component likelihoods of equation (23) are tabulated for every pair on a geometric grid of 1024 values of s between 0.25 and 2.5 by Gauss–Legendre quadrature over the latent speed in a window of $\pm 14\,\sigma_{u,j}$ around u_j, with 128 nodes, and interpolated linearly in the logarithm. The function $\mathcal{A}_j^{\rm good}$ is tabulated once as a function of $x = s/\sigma_{u,j}$ on a logarithmic grid, using trapezoidal integration over $\widetilde{u}$ and the survival function of the non-central χ^2 distribution with two degrees of freedom, $Q_1(a, c) = \Pr[\chi^2_2(a^2) > c^2]$. Values of $\ln \mathcal{A}_j^{\rm good}$ at the grid values of s are obtained by linear interpolation in $\ln x$, and $\mathcal{A}_j^{\rm out}$ is computed once per pair.

Appendix

A Parameters, priors, and posterior computation

The sampled MLR coordinates are

$$\boldsymbol{\theta}_M = \left(c_0, \{a_h\}_{h=2}^{K_Z}, \{b_i\}_{i=2}^{K_x}, \{r_{ih}\}_{i,h\geq 2}, \log\lambda_x, \log\lambda_Z, f_{\text{outlier}}\right), \tag{31}$$

giving 35 scalar degrees of freedom for $K_x = 8$ and $K_Z = 4$. The priors and fixed regularization scales are summarized in table 2. Here $\Pi_{11} = 0.00703$, and the PARSEC-derived reference increments are $\lambda_{x,\text{ref}} = 0.18624$ and $\lambda_{Z,\text{ref}} = 0.13078$ dex.

Table 2: Stage-1a, Stage-1b, and Stage-2 parameter specification. Normal distributions are written as location and standard deviation.

Stage	Quantity	Prior or fixed value	Role
1a	$\boldsymbol{\theta}_{\text{rel}}$	MLE; no prior	Difference calibration (b_G, s_z, ν); fixed in Stage 1b
1b	δ_z	$\mathcal{N}(0, 0.3)$ dex	JCAPS–PARSEC zero point
1b	s_C	Half-Normal(0.1) mag	CMD excess colour scatter
1b	$\boldsymbol{\omega}$	Dirichlet$(1, \ldots, 1)$	Six-component population weights
2	c_0	$\mathcal{N}(\Pi_{11}, 0.10)$ dex	First total-surface coefficient
2	a_h, b_i, r_{ih}	$\mathcal{N}(0, 1)$	Unconstrained monotone coordinates
2	$\log\lambda_x$	$\mathcal{N}(\log 0.18624, 0.7)$	Magnitude increment scale
2	$\log\lambda_Z$	$\mathcal{N}(\log 0.13078, 0.7)$	Metallicity increment scale
2	f_{outlier}	Beta$(3, 12)$	Outlier mixing fraction before selection

Stage 1b and Stage 2 were sampled with the No-U-Turn Sampler (NUTS) (Hoffman and Gelman, 2014), using four chains and 1,000 retained draws per chain. Neither fit had divergent transitions; the largest $\widehat{R}$ was 1.0002 for Stage 1 and 1.0035 for Stage 2, and the smallest effective sample sizes were 2679 and 1291, respectively. Posterior adequacy must additionally be assessed with rank-based $\widehat{R}$, effective sample sizes, chain traces, and posterior predictive or held-out checks; successful execution alone is not scientific validation.

B Information propagated, assumptions, and limitations

Table 3 distinguishes uncertainty that is represented in the current likelihood from uncertainty held fixed or omitted. In particular, the model propagates the reported per-star JCAPS errors, heavy-tailed metallicity scatter, CMD colour scatter, the single-system metallicity posterior, and the reported dynamical amplitude error. It does not propagate uncertainty in the absolute magnitudes, extinction correction, PARSEC model family, age distribution, or uncertainty in the preliminary Stage-1a relative-calibration estimates.

Table 3: Uncertainty and information flow in the model.

Category	Represented	Not represented or deliberately cut
Metallicity	Raw per-star errors, shared Z_j, Student-t tails, CMD excess scatter, population weights, and each system's $\overline{P}_{jq}$	Uncertainty in the fixed preliminary Stage-1a estimates b_G, s_z, and ν; cross-system covariance induced by shared Stage-1b parameters
Photometry and models	Formal BP/RP colour errors and fitted s_C	M_G uncertainty, extinction uncertainty, age as a latent variable, PARSEC-family/model discrepancy
Dynamics	Rice convolution with reported $\sigma_{u,j}$, latent velocity component distributions, and truncation by the cut $u_j > c\,\sigma_{u,j}$	Uncertainty in the fixed good/outlier shapes, selection by the other catalogue criteria, unresolved companions beyond the outlier component, and additional astrometric covariance beyond the reported $\sigma_{u,j}$
Feedback	Metallicity uncertainty is passed from Stage 1b to Stage 2	Dynamics cannot update δ_z, s_C, $\boldsymbol{\omega}$, or individual metallicity posteriors

Several interpretation boundaries follow. First, the shared Z_j is a model assumption for a co-natal binary, not direct proof that both stars have identical surface abundances. Second, the 4000-K threshold expands CMD coverage but may also increase sensitivity to low-temperature PARSEC colour systematics. Third, hard monotonicity encodes the adopted physical ordering of the MLR; it does not demonstrate that PARSEC or the observations are unbiased over the full domain. Fourth, the cut reduces feedback from a potentially misspecified dynamical model, but it understates joint uncertainty by fixing the Stage-1a estimates, discarding the shared Stage-1b posterior covariance, and excluding any information that the dynamics could provide about metallicity calibration. A fully joint analysis would be a different model, not merely a more accurate implementation of the present one.

Two assumptions concern the dynamical term. The truncation of equation (25) models only the threshold equation (24); the remaining catalogue criteria are assumed not to depend on u_j at fixed mass, and $\sigma_{u,j}$ is treated as fixed for each pair. The Stage-1b metallicity posteriors $\overline{P}_{jq}$ are computed from the selected sample and carry no selection correction. The pass probability $\mathcal{A}_j$ is evaluated with the same $p(\widetilde{u})$ and outlier component as the likelihood, so any mismatch between these fixed shapes and the true population affects both $\mathcal{L}_j$ and $\mathcal{A}_j$. The constants in equation (21) come from the weighted simulation described above, which uses a single eccentricity power law at every separation and fits a histogram without parameter uncertainties. The eccentricity index is the only input of the simulation that the sample does not determine. We bracket it by $\alpha_e \in [1.0, 1.5]$: the lower value is the thermal distribution and allows for the upward bias of the measured index from the proper-motion significance cut of Hwang et al. (2022) and for the extrapolation from their sample within 200 pc to ours within 1 kpc; the upper value allows for unresolved companions, which randomize the measured velocity directions and bias the index towards the thermal value. Repeating the full inference with the shapes for $\alpha_e = 1.0$ ($B = 2.2943 \times 10^{-3}$, $u_c = 36.2851$, $C = 3.5567$) and $\alpha_e = 1.5$ ($B = 2.6522 \times 10^{-3}$, $u_c = 37.9042$, $C = 1.7151$) changes the masses by -0.4% to -1.0% and by $+0.5\%$ to $+1.6\%$ over $M_G = 5$–13

at [M/H] = 0, and by the same amounts to within 0.6% at [M/H] = −0.6 and +0.3. This is smaller than the half-width of the 68% credible interval at every M_G. Checks with a fixed shape show that the mass scale of subsamples is stable against projected separation, but it varies with [M/H] and with RUWE, and the predicted tail of the observed u distribution at large separation is lighter than in the data, which the fixed outlier fraction does not follow. The weights of the simulation use only the observed $(s_{\rm proj}, d)$ of the systems, not their u_j, so the u/σ_u threshold enters the model once, through $\mathcal{A}_j$, and not again through the shape. The single shape stands in for the conditional distribution $p(\widetilde{u} \,|\, s_j, d_j)$ of each system. Assuming the separation-dependent index of Hwang et al. (2022) at the observed $s_{{\rm proj},j}$ (5–95% range 1.04–1.34; pooled index 1.234, as adopted), we compared the selection-corrected likelihood with a separate shape for each system and with the single shape, on the masses of the fitted relation. The best-fitting mass scale differs by +0.2% for the whole sample and by +0.8% to −0.2% across quintiles of $s_{\rm proj}$, and $\ln \mathcal{A}_j$ by less than 0.9% for every system, so the approximation is negligible compared with the statistical uncertainty and with the eccentricity systematic above. Building the weights from the 14 876 systems of the likelihood rather than the full catalogue table changes the shape and the masses by less than 0.01%. This approximation therefore cannot account for the trends of the data with separation noted above; these require a different distribution of eccentricities or orbital configurations at small separations, or contamination. The outlier constants in equation (22) are project choices.

C Constructive monotone reparameterization

The total coefficient grid Θ is generated from unconstrained raw variables rather than by rejecting proposals that violate inequalities. Define $\mathrm{softplus}(t) = \log(1 + e^t)$ and $\mathrm{sigmoid}(t) = (1 + e^{-t})^{-1}$. With positive scales $\lambda_x = e^{\log \lambda_x}$ and $\lambda_Z = e^{\log \lambda_Z}$, construct the first row and column as

$$\Theta_{11} = c_0, \tag{32}$$

$$\Theta_{1h} = \Theta_{1,h-1} + \lambda_Z \,\mathrm{softplus}(a_h), \qquad h = 2, \ldots, K_Z, \tag{33}$$

$$\Theta_{i1} = \Theta_{i-1,1} - \lambda_x \,\mathrm{softplus}(b_i), \qquad i = 2, \ldots, K_x. \tag{34}$$

The first row therefore increases with Z, whereas the first column decreases with x. Interior points are visited in increasing $i + h$. For each point, define

$$L_{ih} = \Theta_{i,h-1}, \qquad U_{ih} = \Theta_{i-1,h}, \tag{35}$$

and set

$$\Theta_{ih} = L_{ih} + (U_{ih} - L_{ih}) \,\mathrm{sigmoid}(r_{ih}), \qquad i, h \geq 2. \tag{36}$$

If all previously constructed points obey the two monotone orders, then

$$L_{ih} = \Theta_{i,h-1} \leq \Theta_{i-1,h-1} \leq \Theta_{i-1,h} = U_{ih}. \tag{37}$$

Thus equation (36) lies inside a non-empty interval and simultaneously satisfies $\Theta_{i,h-1} \leq \Theta_{ih} \leq \Theta_{i-1,h}$. Induction over $i + h$ therefore proves the coefficient inequalities in equation (14) for the complete grid. Finally, $D = \Theta - \Pi$ is computed deterministically.

For completeness, the derivative identity for a degree-d_x B-spline surface has the form

$$\frac{\partial g}{\partial x} = \sum_{i=1}^{K_x - 1} \sum_{h=1}^{K_Z} \rho_{x,i} (\Theta_{i+1,h} - \Theta_{ih}) \widetilde{B}_i(x) C_h(Z), \qquad \rho_{x,i} > 0, \tag{38}$$

where $\widetilde{B}_i$ is a non-negative degree-$(d_x - 1)$ basis function. The corresponding Z derivative contains $\rho_{Z,h}(\Theta_{i,h+1} - \Theta_{ih}) B_i \widetilde{C}_h$ with $\rho_{Z,h} > 0$. The coefficient orders therefore establish the derivative signs everywhere in the spline domain, including the appropriate one-sided limits at its boundaries.

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